Band-1 Hadamard Order 7492 — Miyamoto (Paley-I Seed H_1872 q=1871 + C_1874); Miyamoto-only NOT in SPK; Gap 2052 Still OPEN

H_1872 Paley-I PASS; C_1874 PASS; H_7492 Miyamoto PASS; classical SPK FAIL → Miyamoto-only NOT in SPK; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-16

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-16_Band1_Hadamard_7444_Miyamoto`; `2026-09-16_Band1_Hadamard_7396_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_7492_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1872 and 7492 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-I DIRECT seed H_{1872} = Paley-I($q = 1871$) and symmetric Paley conference C_{1874} over GF(1873) ($q = 1873 \equiv 1 \pmod 4$). Classical SPK for 7492: Paley-I **FAIL**s ($q = 7491 = 3\cdot 11\cdot 227$ composite); Paley-II **FAIL**s ($q = 3745 = 5\cdot 7\cdot 107$ NOT prime power); Kronecker **FAIL**s (4×1873). Therefore order **7492 is a Miyamoto-only bonus NOT in SPK**. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under SPK. `clay_closed = false`. Prior ship: **7444 Miyamoto+Paley-II IN SPK**. Prior SPK seeds **1860 / 7444 / 1848 / 7396 / 1788 / 888 / 1776 / 860 / 1720 / 6884 / 1708 / 840 / 1680 / 6724 / ...** retained; Miyamoto-only bonuses **7156/7108/6836/6788/6676/6484/6404/6388/6196/...** and **7492** excluded; **1872** folded into ONBOX/SPK when verified.

Abstract

Finite constructions of orders **1872** and **7492**.

- H_{1872} = Paley-I ($q = 1871$) — preferred seed (Kronecker 4×468 alternate unused)
- C_{1874} = Paley conference over GF(1873), $q = 1873$
- H_{7492} = Miyamoto(H_{1872}, C_{1874})
- Paley-I for 7492: **SKIP** ($q = 7491$ composite)
- Paley-II for 7492: **SKIP** ($q = 3745$ NOT prime power)
- Kronecker for 7492: **SKIP** ($4 \cdot 1873$ not admissible)

SPK: 7492 Miyamoto-only NOT in SPK. Seed 1872 enters SPK (Paley-I). Prior SPK **1860 / 7444 / 1848 / 7396 / 1788 / ...** retained.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_7492.py`

Results. overall_pass=true; units 119/119; wall 343.9779s; all max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 7492 = 4 \cdot 1873$; 1873 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1872} and C_{1874} (Paley conference over prime field $q = 1873$).
- Seed: Paley-I $q = 1871$ (prime $\equiv 3 \pmod 4$) → H_{1872} (Paley-II $q = 935$ FAIL; Double FAIL; Kronecker 4×468 alternate unused).
- SPK: Paley-I $q = 7491$ **FAIL**; Paley-II $q = 3745$ **FAIL** (NOT prime power); Kronecker **FAIL** → **Miyamoto-only NOT in SPK**.
- Prior: 7444 Miyamoto+Paley-II IN SPK shipped; this ship $q = 1873 \rightarrow n = 7492$.

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_1872 Paley-I	Paley-I q=1871 DIRECT	0	4.8331
C_1874	Paley conf GF(1873) q=1873	0	4.6079
H_7492 Miyamoto	Miyamoto(H_1872 Paley-I, C_1874)	0	330.3297
H_7492 Paley-I	SKIP q=7491=$3 \cdot 11 \cdot 227$	—	—
H_7492 Paley-II	SKIP q=3745=$5 \cdot 7 \cdot 107$	—	—
H_7492 Kronecker	SKIP $4 \cdot 1873$ not admissible	—	—

SPK closure note

Seeds include prior SPK ships through **7444** lineage (**1860** Paley-II; **7444** Paley-II GF(61^2); **1848** Paley-I; **7396** Paley-II; **1788**; **888/1776**; **860/1720/6884**; **1708**; **6724**; **840/1680**; **...**; **NOT** 7156/7108/6836/6788/6676/6484/6404/6388/6196) + verified **1872** (Paley-I); **exclude** Miyamoto-only bonuses including **7492**. Known count 1484. Next constructive gap **2052** (OPEN).

Quantity	Value
7492 in SPK	**False** (Miyamoto-only)
1872 in SPK	**True** (Paley-I q=1871)
7444 / 1860 in SPK	**True** (prior)
7156 / 7108 / 6836 in SPK	**False** (Miyamoto-only; excluded)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Non-claims

No full Hadamard conjecture · no Clay-closed · no finished-ToE · no RH / P!=NP / YM / Frankl / NS-solved / AC-solved · gap 2052 remains OPEN · finite constructions of 1872+7492 ONLY · 7492 NOT claimed in SPK.

Repro

Lab: `/workspace/experiments/band1/hadamard_7492_miyamoto/`
Artifacts: `H1872_paleyI_pm1.npz`, `C1874_pm1.npz`, `H7492_miyamoto_pm1.npz`, `results.json`, `literature_ledger.json`, `SUMMARY.txt`.